

22. Which one of the following statements about *electric field strength* and *electric potential* is **incorrect**?

- A Electric potential is a scalar quantity.
- B Electric field strength is a vector quantity.
- C** Electric potential is zero whenever the electric field strength is zero.
- D The potential gradient is proportional to the electric field strength.

Answer: C

- A: Correct. Electric potential is a scalar quantity.
- B: Correct. Electric field strength is a vector quantity.
- C: Incorrect. $E = -\frac{dV}{dx}$, so E is zero if and only if potential gradient is zero
- D: Correct. $E = -\frac{dV}{dx}$